



Dissection of mechanisms of Chinese medicinal formula Si-Miao-Yong-an decoction protects against cardiac hypertrophy and fibrosis in isoprenaline-induced heart failure



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ABSTRACT

Ethnopharmacological relevance: Si-Miao-Yong-An decoction (SMYAD) is a traditional Chinese herbal formulation. SMYAD first appeared in the Eastern Han Dynasty according to the “Shen Yi Mi Zhuan”. Then the formula was recorded in the “Yan Fang Xin Bian” edited by medical scientist Bao Xiangao in the Qing Dynasty. This well-known prescription has been traditionally used for gangrene and vascular vasculitis. It is mainly used for cardiovascular and endocrine diseases in current clinical applications and research.

Aim of study: In this study, the potential mechanisms of SMYAD against cardiac fibrosis and hypertrophy in the β -adrenoceptor agonist isoprenaline induced heart failure model were investigated.

Materials and methods: The heart failure animal model was established via injected isoprenaline in rats. Echocardiography was used to detect the structure and function of the heart. HE staining and Masson's trichrome staining was performed to assess myocardial tissue morphology. The serum biochemical indexes were detected by dedicated biochemical kit. BNP was tested by ELISA kit. The levels of mRNA were detected by RT-qPCR. Cardiomyocyte morphology was assessed by immunofluorescence. Phosphorylated and total p38, Akt were analyzed by Western blot. The production of reactive oxygen species (ROS) was tested by CM-H2DCFDA probe. Formula identification of chemical constituents of SMYAD in plasma was disclosed through ultra-performance liquid chromatography coupled to quadrupole time-of-flight mass spectrometry (UPLC-Q/TOF-MS).

Results: SMYAD was able to improve the heart function in ISO induced heart failure rat model via protecting rat from developing cardiac hypertrophy and fibrosis. SMYAD also decreased plasma expression of these biochemical indexes. It was found that SMYAD could regulate cardiac hypertrophy and fibrosis makers' mRNA levels *in vitro* and *in vivo*. In addition, SMYAD inhibited the phosphorylation of p38 and Akt, which are key mediators in the pathological process of ISO-induced cardiac hypertrophy and myocardial fibrosis. It also showed that the components of SMYAD in rat plasma exerted myocardial cell protective activity.

Conclusion: In summary, SMYAD may comprise more than one active ingredient to the pursuit of combination therapies instead of specifically target a single disease-causing molecule. These experimental results suggest that SMYAD may be a potential drug candidate in diseases of cardiac hypertrophy and myocardial fibrosis caused by β -adrenoceptor abnormalities.

Abbreviations: Traditional Chinese medicine, (TCM); HF, heart failure; ISO, isoprenaline; EF%, ejection fraction; FS%, fractional shortening; LVESd, left ventricular end-systolic diameter; LVEDd, left ventricular end-diastolic diameter; AST, aspartate aminotransferase; LDH, lactate dehydrogenase; CK, creatine kinase; CK-MB, creatine kinase isoenzymes; α -HBDH, α -hydroxybutyrate dehydrogenase; BNP, brain natriuretic peptide; ANP, atrial natriuretic peptide; β -MHC, β -myosin heavy chain; UPLC-Q/TOF-MS, ultra-performance liquid chromatography coupled to quadrupole time-of-flight mass spectrometry; ROS, reactive oxygen species

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1. Introduction

Heart failure (HF) is the last pathological process of many cardiovascular diseases. Ischemic heart disease, hypertension, fibrosis, smoking, obesity, and diabetes, these risk factors are the main cause of HF (Krum and Abraham, 2009; Mozaffarian et al., 2016; Kristen et al., 2015). Briefly, due to the contractile function and/or diastolic dysfunction of the heart, the arterial system has insufficient blood perfusion which led to cardiac dysfunction syndrome (Hopper and Easton, 2017; Teerlink and Metra, 2013).

Traditional Chinese medicine (TCM) has been applied successfully for thousands of years and is also widely prescribed in China and some Asian countries. TCM differs from western medical theory system in terms of multiple targeting and systemic metabolic regulations (Sun, 2015). Si-Miao-Yong-An decoction (SMYAD) comprises Flos Lonicerae Japonicae (*Lonicera japonica* Thunb., Jinyinhua), Radix Scrophulariae Ningpoensis (*Scrophularia ningpoensis* Hemsl., Xuanshen), Radix Angelicae Sinensis (*Angelica sinensis* (Oliv.) Diels, Danggui) and Radix Glycyrrhizae Uralensis (*Glycyrrhiza uralensis* Fisch., Gancao) in a 3:3:2:1 proportion by weight (Liu et al., 2017). This famous ancient recipe has the efficacy of clearing away heat and detoxifying, promoting blood circulation and removing blood stasis (Jiang, 2004). Therefore, SMYAD was used to treat dislocation in ancient times according to the traditional Chinese medicine classics (Yan Fang Xin Bian) (Lin et al., 2012). Due to the characteristics and efficacy of SMYAD, it was used clinically to treat diseases such as thromboangiitis obliterans, coronary heart disease, angina pectoris, hyperlipidemia, hypertension, viral myocarditis and other diseases nowadays (Wang et al., 2018; Wu, 2015).

Recently, studies have reported that SMYAD could reduce fat accumulation in hyperlipidemia rat liver to improve hyperlipidemia, which is considered to be a major risk for atherosclerosis, cardiovascular disease (CVD) (Liu et al., 2017; Li et al., 2017; Ma et al., 2017). In addition, a previous study also revealed that SMYAD has beneficial effect against atherosclerotic plaque in rabbit model (Peng et al., 2012). Moreover, SMYAD could confront against matrix degradation, lipid deposition and inflammatory to promote atherosclerotic plaque stability (Peng et al., 2010; Yu et al., 2016b). It has been shown that SMYAD protect human umbilical vein endothelial cells from hypoxia injured and promoting angiogenesis (Liu et al., 2010). Moreover, *Lonicera japonica* has shown potent myocardial protective activity against the ischemic heart diseases related to oxidative stress (Wang et al., 2016; Jeong et al., 2009). It is reported that extracts from Radix Scrophulariae attenuated ventricular remodeling, which is considered as the basic mechanism in the process of heart failure (Huang et al., 2012). In addition, some flavonoids and polysaccharide from *Angelica sinensis* of SMYAD could attenuate atherosclerosis, against cardiac hypertrophy and protects H9c2 cells from oxidative injury (Cheng et al., 2017; Huang et al., 2014; Liao et al., 2014; Ma et al., 2016; Niu et al., 2018). Accordingly, the cardioprotective activity of the extracts and compounds from SMYAD might be the material basis of SMYAD in cardiovascular disease. The pharmacokinetic study of the above components would help to elucidate the mechanism of SMYAD and provide reference for other research of TCM.

Therefore, based on the pharmacodynamics study of SMYAD, we further investigated the efficacy of SMYAD on complex heart diseases. In this study, water extracts of SMYAD were prepared, and the effect of against cardiac hypertrophy and fibrosis in isoprenaline-induced rat heart failure model were examined to explore the potential therapeutic effects of SMYAD on complex cardiovascular diseases.

2. Materials and methods

2.1. Reagents and materials

All herbs were purchased from Changshengtang Pharmacy (Shenyang, China), in Sep 10, 2016, and authenticated by Prof. Jiuzhi

Yuan, at Shenyang Pharmaceutical University, China. The voucher specimens (*Lonicera japonica* Thunb, No. ZYQLJ-2016; *Scrophularia ningpoensis* Hemsl., ZYQSN-2016; *Angelica sinensis* (Oliv.) Diels, ZYQAS-2016; *Glycyrrhiza uralensis* Fisch, ZYQGU-2016) have been deposited at College of Traditional Chinese Materia Medica, Shenyang Pharmaceutical University, China. *Lonicera japonica* Thunb, *Scrophularia ningpoensis* Hemsl., *Angelica sinensis* (Oliv.) Diels and *Glycyrrhiza uralensis* Fisch. were weighted in a proportion of 3:3:2:1. Then the mixture was extracted with water (1:8, w/v) twice under refluxing, 2 h per time. After the extraction solutions were mixed, they were filtered and concentrated by a rotary evaporator. The final extract was concentrated to 3 g/mL which means 3 g mix crude herbs' extract to 1 mL water.

Isoprenaline (I5627-5G) was purchased from Sigma-Aldrich (St. Louis, MO, USA). The rat BNP ELISA Kit (ab108816) was obtained from Abcam (Cambridge, UK). DMEM medium, Trypsin, penicillin, streptomycin and fetal bovine serum purchased from Hyclone (St. Louis, MO, USA). MTT (M2128), DMSO, 2, 2, 2-tribromomethane, DCFH-DA (D6883) and DAPI (D9542) were purchased from Sigma (St. Louis, MO, USA). RIPA lysate (strong), PMSF, ECL chemiluminescence kit was purchased from Beyotime (Beijing, China). BCA protein quantitative kit was purchased from Biosharp (Beijing, China). Phospho-p38 MAPK (Thr180/Tyr182), (CST, #4511), p38 MAPK, (CST, #8690), Akt (pan), (CST, #4691), phospho-akt (Thr308), (CST, #13038), GAPDH (CST, #5174), actin (CST, #3700) and other primary and secondary antibodies labeled with horseradish peroxidase were purchased from Cell Signaling Technology (CST, Beverly, MA).

2.2. Animals and experiment protocols

Male Sprague-Dawley (SD) rats (male, 200–220g) were purchased from Beijing Hua Fukang Biotechnology Company Limited. All animals were raised in a 12 h light/dark cycle room, which is in a state of humidity at $40 \pm 5\%$ and temperature at $22\text{--}24^\circ\text{C}$. These rats were free to standard diets and drinking water and received humane care. The animal experiments were given official approval by Animal Care and Use Committee of Shenyang Pharmaceutical University. The animal experiments were also complied with the Guide for the Care and Use of Laboratory Animals, published by the US National Institute of Health (NIH publication no. 85–23, revised in 1996).

The healthy male rats were chose for experimentation. There are CON group ($n = 10$) and the other groups. The other rats were subcutaneously injected with 5 mg/kg/d ISO for one week. And then, echocardiogram was performed on rats to evaluate the heart failure model. After successful establishment of the model, CON group and ISO were fed with physiological saline 3mL/kg/d. In SMYAD (L) + ISO, SMYAD (M) + ISO and SMYAD (H) + ISO group, the animals were intragastric administration with SMYAD 10 g/kg/d, 20 g/kg/d and 40 g/kg/d, respectively. After four weeks, the indicators mentioned in the article were evaluated, including echocardiography, histopathology examinations and the heart weight index.

2.3. Echocardiography/Hemodynamic parameters measurements

At 4-week, anesthetized experimental animals with 2, 2, 2-Tribromethanol (200 mg/kg, Sigma) and the cardiac function was assessed via echocardiography using an Acuson Sequoia C256 (Siemens) ultrasound machine. Fractional shortening (FS%), ejection fraction (EF %), left ventricular end-diastolic diameter dimensions (LVEDd) and left ventricular end-systolic diameter (LVESd) were detected as previously reported (Liu et al., 2009). The rat heart weight and left ventricular weight were calculated after dissection of the rats.

2.4. Histology examination

Freshly heart tissue was embedded in OCT embedding matrix

(Thermo Fisher Scientific) and stored in -80°C freezer. Sections were cut into 5 or 10 μm thickness cryosections. Haematoxylin and eosin (HE) staining and Masson's trichrome staining were carried out. (Liu et al., 2011).

2.5. Analysis of serum biochemical indexes

The levels of lactate dehydrogenase (LDH), aspartate aminotransferase (AST), creatine kinase isoenzymes (CK-MB), creatine kinase (CK) and α -hydroxybutyrate dehydrogenase (α -HBDH) in plasma were measured by automatic chemistry analyzer (Mindray BS-120; Mindray Bio-Medical Electronics Co., Ltd., Shenzhen, China). BNP (abcam, ab108816) was detected by enzyme-linked immunosorbent assay and operated according to the experimental instructions (Cowie and Mendez, 2003; Hall, 2014).

2.6. Cell lines and culture

Rat myoblast cell line H9c2 was purchased from American Type Culture Collection. H9c2 was cultured in DMEM Low Glucose with 10% fetal bovine serum (Gibco) at 37°C with 5% CO_2 in a humidified incubator.

2.7. Cell viability assay (MTT assay)

Cell viability was assessed by addition of 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyl-tetrazolium bromide (MTT) according to the manufacturer's protocol. 5000 cells were seeded per well in 96-well plates. After incubation for 24 h, the cells were added with different concentrations of drug-containing medium for 24 h at 37°C with a 5% CO_2 atmosphere. Add 5mg/ml MTT per well and the plates were incubated for 1–4 h. Discard the medium and add 150 μL of DMSO to dissolve the crystal formazan dye. Finally, the absorbance was quantified at wavelength of 570 nm using the InfiniteF500 micro-plate reader (Tecan). All assays were performed in triplicate, and the data were normalized to DMSO treated control and analyzed using Prism software (GraphPad Software).

2.8. RNA extraction and RT-qPCR

Weigh a certain amount of cardiac tissue, homogenate and extract total RNA using TRIZOL kit (Invitrogen, Carlsbad, CA). Cellular RNA was extracted using the same method. Reverse transcription experiments were performed using the kit TransScript All-in-One First-Strand cDNA Synthesis SuperMix for RT-qPCR (One-Step gDNA Removal) (Transgene, AT341). Real-time fluorescent quantitative PCR was performed via iQ[™] SYBR[®] Green Supermix (Bio-Rad) kits and was carried out in accordance with the manual operation. The primer sequences used in the experiment are shown in Table 1. The level of each gene was normalized by the amount of GAPDH. RT-qPCR was performed as previously reported (Tovar et al., 2006).

2.9. Western blotting

Cardiac tissue was weighed and homogenized, and then RIPA lysate in the presence of Protease Inhibitor Cocktail (100X) (CST, #5871) was

added. After plating and incubating the H9c2 cells for 24 h, the cells were collected sampled after treatment with ISO and SMYAD according to the experimental requirements. The samples were lysed with RIPA buffer that including Protease Inhibitor Cocktail (100X). Then the samples were incubated on ice until fully lysed and then centrifuged at 12,000g for 15 min at 4°C . Leave the supernatant. The protein concentration of each sample was calculated with the BCA protein quantitative kit (CWBIO, Beijing). Denatured the protein extract and then run on SDS-PAGE gels with 30 μg protein for each sample. After electrophoresis, the proteins were transferred to PVDF membrane (Milipore, ISEQ08100). Blocked the membrane for 1 h at room temperature with 5% BSA blocking buffer (5% BSA in TBS/0.1% Tween-20). Incubated the membrane with primary antibodies overnight at 4°C . Wash the membrane three times with TBS/0.1% Tween-20. And secondary infrared antibodies goat anti rabbit IgG (Proteintech, 1 : 5000), goat anti mouse IgG (Proteintech, 1 : 5000) were added. After washed with TBST for three times, the Immobilon Western Chemiluminescent HRP Substrate (Milipore, WBKLS0050) was used and then scanned the Blots on a Bio-rad gel imaging system (SYSTEM GelDoc XR+).

2.10. Immunofluorescence staining

Briefly, cells were seeded on glass coverslips. After adhesion, incubation was conducted with the SMYAD (80 μM) or SMYAD (80 μM) + ISO (10 μM). After 36 h, the cells were fixed with 4% paraformaldehyde. It was then permeabilized in 0.15% Triton X-100 for 20 min and saturated with 4% BSA in PBS for 1 h at room temperature. Then the cells were incubated overnight with β -Actin (CST, #3700) at 1:200 dilutions in PBS-4% BSA following incubation with a FITC-conjugated secondary antibody (ab6717). Slides were observed via a Nikon A1RMP + positive two-photon confocal microscope.

2.11. ROS detection

ROS levels were detected via CM-H2DCFDA probe (Invitrogen). H9c2 cells were treated with 10 μM of Doxorubicin (DOX) in the presence or absence of SMYAD for 6 h and then stained with CM-H2DCFDA. Images were observed and captured via Zeiss OBSERVER D1/AX10 cam HRC inverted fluorescence microscope. Adobe Photoshop 7 was used only for brightness and contrast. Next, flow cytometry was used to detect the positive rate of ROS after treatment and staining. Data were obtained in BD FACSAria II cytometry.

2.12. Statistical analysis

The statistical significance of the data between groups was acquired either Student's t-test (for two groups) or one-way ANOVA multiple comparisons in GraphPad Pro 7.0 (GraphPad, San Diego, CA). P values < 0.05 ($p < 0.05$) were considered statistically significant. All experiments were repeated at least three times.

Table 1
Primers sequences used for real-time PCR reactions.

Gene	Forward primer	Reverse primer
ANP	GGAGCCTGCGAAGGTCAA	TATCTTCGGTACCGGAAGCTGT
B-MHC	AGCTCCTAAGTAATCTGTTGCCAA	AAAGGATGAGCCTTTCTTTGCT
Collagen	TGTTGACATTGCACCCATGG	CAGCCATTCAATGCAGATCC
SERCA2a	TGAGATAGCGGGCATACTGG	AGTTCCTGGTTCTTCTCCCG
GAPDH	CAAGTTCAACGGCACAGTCA	CCCCATTGTATGTTAGCGGG

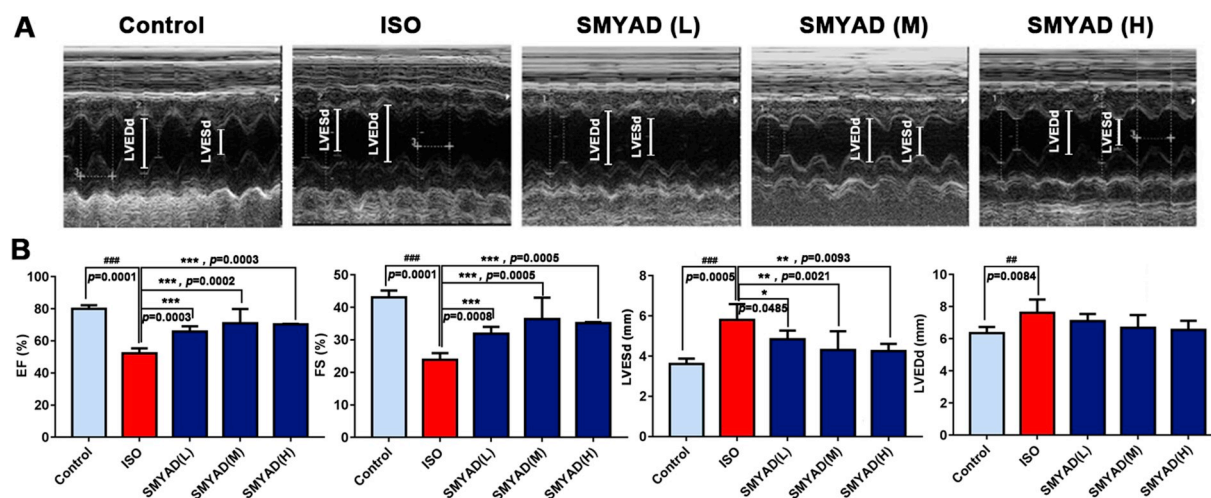


Fig. 1. SMYAD recovers cardiac function in ISO-induced heart failure. (A) Representative images of echocardiography (B) Echocardiographic assessment of EF%, FS %, LVESd and LVEDd by quantification evaluated the impaired cardiac function (n = 6–9). *** $P < 0.001$, ** $P < 0.01$, * $P < 0.05$ compared with ISO group, ### $P < 0.001$, ## $P < 0.01$, # $P < 0.05$ compared with control.

3. Results

3.1. Evaluate of cardiac function by echocardiographic assessment

Echocardiographic assessment of ejection fraction (EF%), fractional shortening (FS%), left ventricular end-systolic diameter (LVESd), and left ventricular end-diastolic diameter (LVEDd) by M-mode images were executed to reveal the changes of cardiac function. As shown, the cardiac contractile function was notably reduced in ISO induced heart failure rat model, exemplified by decreased EF% and FS%, increased LVESd and LVEDd. However, after 4-weeks treatment, LVESd and LVEDd were lower in SMYAD groups than the model group. As a tendency, EF% was increased. And an increase of FS% was also detected compared with model group (Fig. 1A and B). These results demonstrated that SMYAD improved the cardiac function and enhanced the capacity to withstand HF stress induced by ISO.

3.2. SMYAD protects rat from developing cardiac hypertrophy and fibrosis

As shown in Fig. 2A, compared with the ISO group, the heart weight index (HWI) and left ventricular weight index (LVWI) were reduced in the SMYAD group. The results demonstrated that the degree of cardiac hypertrophy in the SMYAD group was reduced. In addition, SMYAD treatment declined that ISO-induced accumulation of fibrosis in the myocardial space interstitial space and the increased cardiomyocytes size as disclosed by HE and Masson's trichrome staining (Fig. 2B and C).

3.3. Plasma expression of LDH, AST, CK, CK-MB, α -HBDH, BNP

LDH, AST, CK, CK-MB and α -HBDH have been used as biomarkers for diagnosing myocardial injury (Yu and Wang, 2017; Liu et al., 2015b). And BNP have diagnostic utility in the detection of left ventricular hypertrophy and left ventricular systolic dysfunction (Sato et al., 2018). ISO treatment could induce myocardial injury which may ultimately lead to myocardial necrosis and left ventricular hypertrophy. Subsequently, myocardial injury caused the release of LDH, AST, CK, CK-MB, and α -HBDH into bloodstream. Meanwhile, the plasma concentration of BNP was increased in chronic congestive heart failure. In this study, a significant increase of the above biomarkers was detected in ISO group, while decreased plasma expression of these biomarkers was examined in SMYAD groups (Fig. 3). The results further illustrated that the extracts of SMYAD protected rats against the ISO-induced myocardial injury and left ventricular hypertrophy.

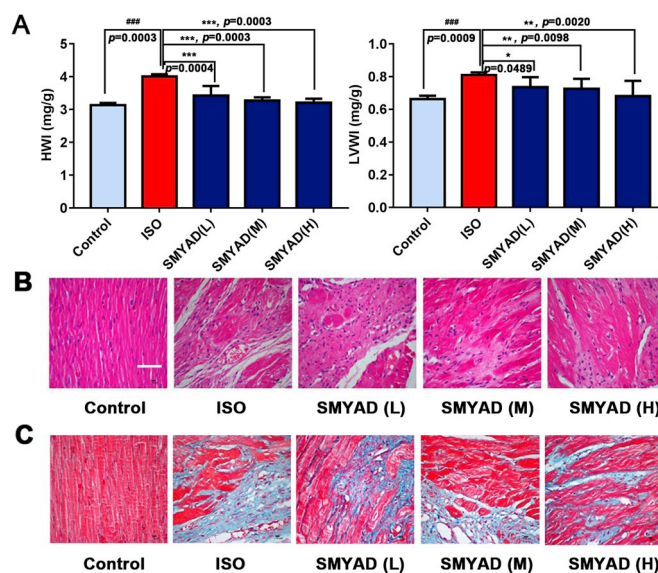


Fig. 2. Histological analysis of cardiac hypertrophy and myocardial fibrosis. (A) Ratio of heart weight to body weight. HWI, heart weight index; LVWI, left ventricular weight index. (B) HE staining results of cardiac tissue represented. (C) Masson staining results of cardiac tissue represented. Heart tissues were visualized via a light microscope at $\times 400$ magnification. Data are presented as means $\pm$ SD, n = 8. *** $P < 0.001$, ** $P < 0.01$, * $P < 0.05$ compared with ISO group, ### $P < 0.001$, ## $P < 0.01$, # $P < 0.05$ compared with control.

3.4. SMYAD attenuates pre-established cardiac hypertrophy and fibrosis in rat

It has been reported that SMYAD possess anti-hyperlipidemia and ease atherosclerosis activities (Liu et al., 2017; Peng et al., 2012). Here, the other effects of SMYAD on cardiovascular diseases were investigated. In this study, we found that SMYAD treatment reduced the activation of the interstitial fibrosis gene collagen (Fig. 4A). The transcription levels of myocardial hypertrophy makers were also decreased. In ISO-induced rat heart failure model, the excitatory contraction coupling abnormalities mainly due to myocardial cell calcium dysregulation (Shah and Mann, 2011). Calcium dysregulation was correlated with arrhythmia and diastolic dysfunction. More important, Ca^{2+} signals are the mediators of gene expression changes in cardiac hypertrophy response of cardiomyocytes (Zhang and Brown, 2004). Ca^{2+}

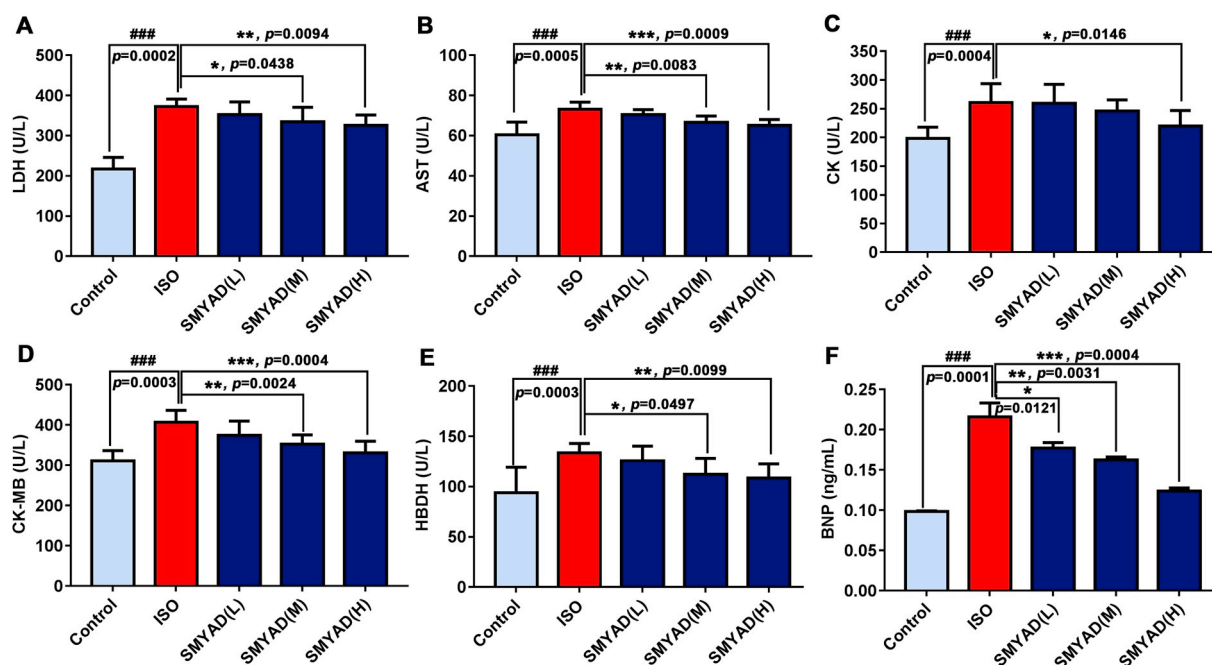


Fig. 3. Analysis of serum biochemical indexes. A-F Serum AST, LDH, CK, CK-MB, α -HBDH and BNP level were determined using enzymatic kits. Data are presented as means $\pm$ SD, n = 8. *** P < 0.001, ** P < 0.01, * P < 0.05 compared with ISO group, ### P < 0.001, ## P < 0.01, # P < 0.05 compared with control.

ATPase pump (SERCA2a, also known as ATP2A2) is a key factor in Ca^{2+} cycle regulation. Therefore, enhancement of SERCA2a activity through increasing the amount of protein has been used in targeting impaired cardiac contractility clinical heart failure treatment (Jaski et al., 2009). As shown in Fig. 4A, SMYAD increased SERCA2a transcription levels, which indicated the improvement of the cardiac contractility.

Then the signaling pathways associated with cardiac hypertrophy were analyzed. It is known that increased Akt activation could induce cardiac hypertrophy to make a response to variety of stresses (Sussman et al., 2011). Therefore, the role of SMYAD in regulating Akt signaling was examined in the current study. In model group, increased phosphorylation of Akt level was observed. Interestingly, SMYAD decreased the phosphorylation of Akt level in treatment group. Consistent with Akt, p38 was also activated in model group, while its activation was blocked in SMYAD groups (Fig. 4B). p38 MAPK is an important signal pathway that will be activated when myocardial injury accentuated, cardiac fibrosis and diastolic dysfunction (Jeong et al., 2012). In addition, apoptosis can be triggered by activation of p38 which is related to the accumulation of reactive oxygen species (Shah and Mann, 2011). Overall, above data indicated that the remedy of SMYAD in reversing cardiac hypertrophy, improving cardiac fibrosis and recovering cardiac function.

3.5. SMYAD protects H9c2 cardiomyocytes against injury and cardiac hypertrophic induced by ISO

Based on the above results, myocardial protective effect of SMYAD was next studied *in vitro*. As shown in Fig. 5A, SMYAD treatment improved the viability in ISO-treated cells in a concentration dependent manner. In addition, the results demonstrated that SMYAD did not interfere with cell survival viability of H9c2 cells up to a concentration of 300 $\mu\text{g/mL}$ (Fig. 5A). Through the UPLC-Q/TOF-MS results, (Fig. 1S, Table 2), 16 compounds were identified or tentatively elucidated in the SMYAD treatment groups. Furthermore, most of the analytes exhibited cardiomyocytes protective activity in the isoproterenol injury model. (Fig. 2S). The absorbed chemical components in blood may exert a synergistic effect to achieve the myocardial protection therapeutic

benefit.

We also found that the mRNA level of ANP and β -MHC were increased, which means the underway of cardiac hypertrophic, while the level were reduced in SMYAD-treated cells. SMYAD treatment also reduced the activation of the interstitial fibrosis maker collagen and increased the SERCA2a mRNA levels, which were consistent with the results *in vivo* (Fig. 5B). SMYAD treatment relieved the ISO-mediated increasing of cardiomyocytes size (Fig. 5C). Moreover, SMYAD suppressed the excessive activation of AKT and p38 signal pathways induced by ISO (Fig. 5D). The above results indicated that SMYAD is capable of protecting H9c2 against injury and blocking the cardiac hypertrophic and myocardial fibrosis response *in vitro*.

3.6. Protective effects of SMYAD on DOX-induced oxidative damage in H9c2 cardiomyocyte cells

DOX could induce acute and chronic heart failure through oxidative damage (Du et al., 2012). Here, the anti-HF effects under oxidative stress of SMYAD were evaluated, which are revealed by the protective effects and the oxidation resistance. As shown, SMYAD treatment increased the cell viability of DOX-treated cells in a concentration dependent manner (Fig. 6A). Elevation of oxidative stress upon DOX treatment was observed by CM-H2DCFDA flow cytometry and fluorescence staining. However, the ROS level was decreased to normal range in DOX plus SMYAD group. These data indicated that SMYAD could protect cardiomyocyte cells from ROS generation to avoid oxidation damage (Fig. 6B and C).

4. Discussion

In the present study, novel observations are provided to evaluate the effect of SMYAD in ISO-induced heart failure animal model. We found that SMYAD treatment attenuated the beta receptor agonist-induced cardiac hypertrophy and hypertrophic response of cardiomyocytes *in vivo* and *in vitro*. Furthermore, SMYAD treatment attenuated its transformation into myofibroblasts and improved the symptoms of myocardial fibrosis. Moreover, SERCA2a expression was also restored in failing hearts, which means to improve abnormal contractions via Ca^{2+}

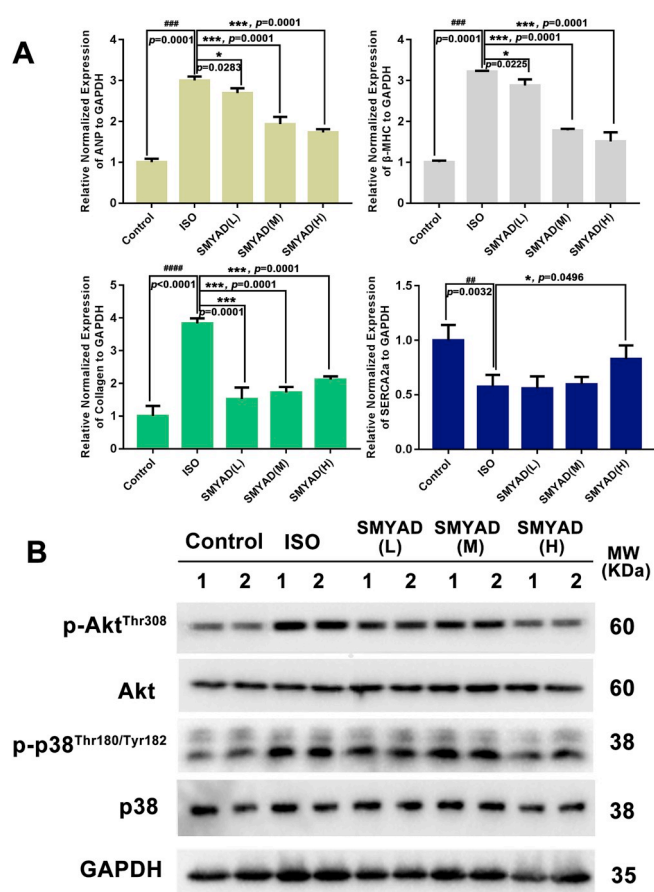


Fig. 4. SMYAD attenuates pathological symptom of heart failure in ISO rat model.

(A) Collagen-1 and SERCA2a mRNA levels of the five groups. (B) Western blot analysis of different groups of rat heart tissue. Results are shown for two animals of each group. Data are presented as means $\pm$ SD, $n = 3$. **** $P < 0.0001$, *** $P < 0.001$, ** $P < 0.01$, * $P < 0.05$ compared with ISO group, ### $P < 0.001$, ## $P < 0.01$, # $P < 0.05$ compared with control.

regulation. Taken together, the results support the claim that SMYAD could alleviate the damaging effects of myocardial hypertrophy and myocardial fibrosis, and functioned as a cardio-protective role in failing heart.

HF is a complex clinical syndrome in etiology and manifestation, which arises due to abnormalities of cardiac structure and blood pumping. The cardiac often entails complex remodeling which performed original physical and morphological changes, while it is an adaptation of the body to such risk factors. It may eventually develop to systolic dysfunction, ventricular dilatation, and arrhythmias (Shah and Mann, 2011; Maclellan et al., 2012). Cardiomyocyte hypertrophy is the prominent feature of remodeling heart. Calcium (Ca^{2+}) handling, contractile function and cell viability also have successive changes (Mudd and Kass, 2008). In the current work, the cardio-protective effect of SMYAD treatment on ISO-induced cardiac hypertrophy was investigated. The results demonstrated that SMYAD decreased the ISO-mediated activation of the hypertrophic markers ANP and β -MHC, fibrosis marker collagen and increased the heart to body weight ratio. Combined with the histopathological section results, we can conclude oral administration of SMYAD could reduce hypertrophic and fibrosis. The results of echocardiography and plasma biochemical makers' detection illustrated that the electrical physiology of cardiomyocytes of calcium (Ca^{2+}) handling should be affected to improve systolic dysfunction. Since it has been reported that isoprenaline significantly reduced SERCA2a to disturb Ca^{2+} homeostasis, the SERCA2a transcriptional level was detected in this study. And it demonstrated that SMYAD

treatment improved the SERCA2a function both *in vitro* and *in vivo*.

Recently, studies have suggested that p38 regulates cellular response to cell growth, apoptosis and various stresses (Khaper et al., 2017) and is involved several organs such as renal fibrosis (Ma et al., 2014) and cardiac fibrosis (Li et al., 2011). In the model of ISO-induced cardiac fibrosis, p38 MAPK was also crucially involved (Lu et al., 2014). SMYAD weakened ISO-induced cardiac fibrosis via suppressing the hyper-activation of p38 in this study. On the other hand, SMYAD also participated in the inhibition of Akt signaling pathway, which implicated in cardiac hypertrophy such as angiotensin II, α -adrenoceptor stimulation.

Through the UPLC-Q/TOF-MS results (Table 2), 16 compounds were identified or tentatively elucidated in the SMYAD treatment groups. Chlorogenic acid and homologous compounds are the active ingredients in *Lonicera japonica*, which have been shown anti-oxidative activity (Xiang and Ning, 2008) and protect H9c2 cell against H_2O_2 -induced apoptosis (Yu et al., 2016a). Previous studies have revealed that angoroside C contributes to improve ventricular remodeling (Gu et al., 2015) and also possess anti-inflammatory activities (Liu et al., 2015a). Besides, quinic acid derivatives work as antioxidant constituents could repress the development and progression of atherosclerotic disease (Hung et al., 2006; Kweon et al., 2004). Furthermore, oxidative stress is an important contributor to myocardial fibrosis, which is damaged with the explosion of ROS production (Purnomo et al., 2013). Isoprenaline as β_2 -adrenoceptor could activate enhancement of ROS production and then activate p38 MAPK in the processes of interstitial fibrosis (Xu et al., 2011; Zhang et al., 2005). Therefore, we hypothesized that cardiac function protective effects of SMYAD in isoprenaline-induced heart failure model may partly depend on the removal of ROS. To confirm this hypothesis, the oxidative stress damage cardiomyocytes cell model was used. The results showed that SMYAD could cope with ROS and reduce the occurrence of oxidative damage.

Besides, we found that SMYAD did not exhibit obvious systemic toxicity when it worked which was observed on the basis of weight. Compared with the model group, the weight of rats in SMYAD group increased and there was no significant difference between the high dose group and the other two groups. This means that SMYAD is effective without obvious systemic toxicity. AST is usually available as an indicator of liver injury except as a cardiovascular disease index (Shin et al., 2008). The result of AST suggested that SMYAD had no hepatotoxicity in high dose group. However, there were still some several limitations in our study. First, the change of these pathways was not confirmed as the unique or direct effect of SMYAD in this study. To evaluate the direct effect, specific inhibitors or RNA interference techniques of each pathway could be used to evaluate the direct effect for future investigation. In addition, given the multiple components in SMYAD, there should be other anti-cardiac hypertrophy and fibrosis mechanisms involved. The other precise mechanisms need to be assessed in further studies.

5. Conclusions

In conclusion, the current study demonstrated that SMYAD prevented isoprenaline-induced cardiac hypertrophy and fibrosis through inhibiting Akt and p38 MAPK pathways. Besides, the antioxidant agents of SMYAD could relieve oxidative damage to improve myocardial fibrosis in cardiomyocytes cells. Furthermore, most of the analytes were shown to be stable in rat plasma samples and exhibited biological activity in isoprenaline induced myocardial damage cell model. These findings suggest SMYAD could be a potential candidate drug against cardiac hypertrophy and fibrosis.

Author contributions

Yuqian Zhao performed data analysis and wrote the paper. Xinsheng Yao and Lingjuan Zhu designed and coordinated the research. Yuqian

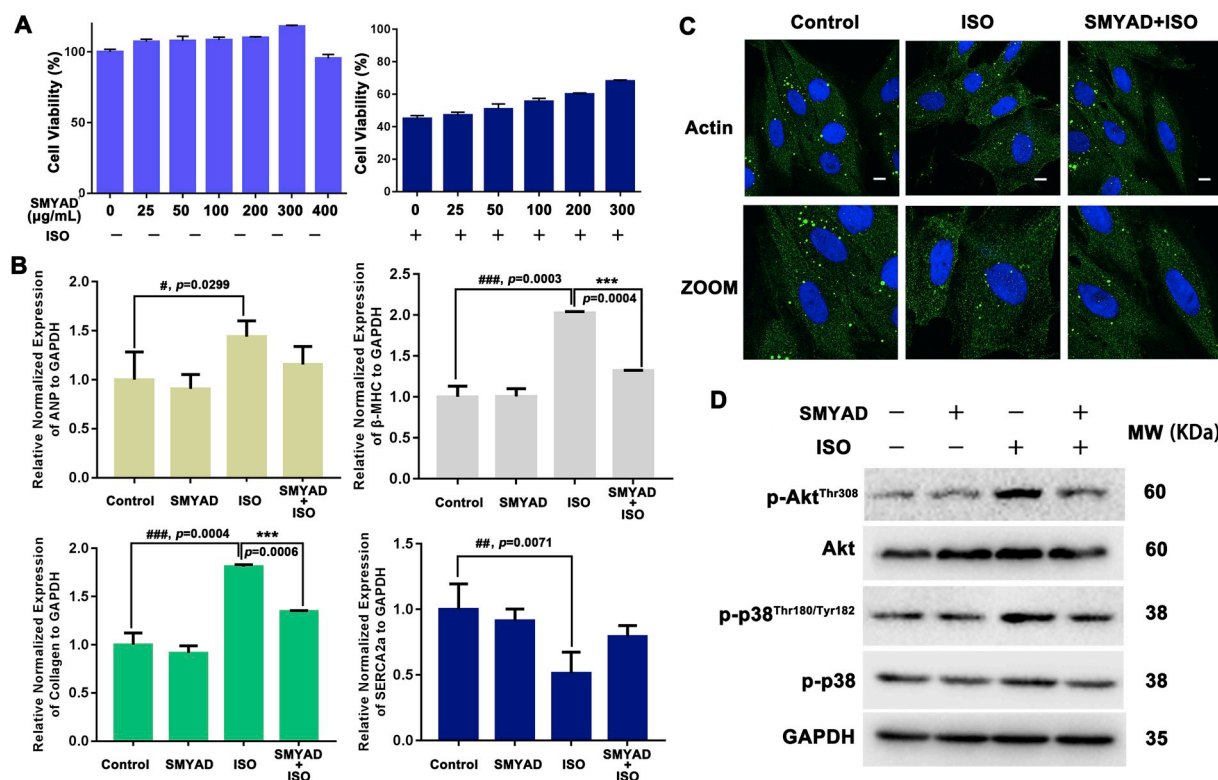


Fig. 5. SMYAD attenuates cardiac hypertrophy and myocardial fibrosis *in vivo*. (A) The effect on cell viability of SMYAD (25–300 µg/mL) was evaluated in H9c2 cells and ISO (80 µM) treatment H9c2 cells. (B) Expression levels of ANP, β-MHC, mRNA, collagen-1 and SERCA2a in different group as indicated. (C) H9c2 was treated with ISO in the presence or absence of SMYAD (300 µg/mL). Cardiomyocyte size was identified by β-actin staining (green). The position of nuclei was marked by DAPI staining. Scale bars, 25 mm. Data are representative of three independent experiments (mean ± SD). ****P* < 0.001, ***P* < 0.01, **P* < 0.05 compared with isoprenaline, ###*P* < 0.001, ##*P* < 0.01, #*P* < 0.05 compared with control. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Zhao and Yingnan Jiang performed the molecular biology experiments. Yanmei Chen performed the animal experiments. Xue Zhang and Fengxiang Zhang performed the UPLC-HR-ESI-MS analysis. All authors revised the paper and approved the final manuscript.

Declaration of competing interest

The authors declare no conflict of interest.

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Table 2

Formula identification of chemical constituents of SMYAD in plasma.

Peak	t _R (min)	Selectedion	Calculated mass (m/z)	Formula	Compound
1	1.43	[M-H] ⁻	353.0873	C ₁₆ H ₁₇ O ₉	3-caffeoylquinic acid
2	1.81	[M-H] ⁻	165.0552	C ₉ H ₉ O ₃	<i>p</i> -hydro-phenylpropionic acid
3	2.17	[M-H] ⁻	353.0873	C ₁₆ H ₁₇ O ₉	5-caffeoylquinic acid
4	2.41	[M-H] ⁻	353.0873	C ₁₆ H ₁₇ O ₉	4-caffeoylquinic acid
5	3.73	[M-H] ⁻ [2M-H] ⁻ [M+H] ⁺ [M+Na] ⁺	403.1240 807.2644 405.1396	C ₁₇ H ₂₃ O ₁₁ C ₃₄ H ₄₇ O ₂₂ C ₁₇ H ₂₅ O ₁₁	Secoxyloganin
6	4.37	[M-H] ⁻ [M+H] ⁺	427.1216 536.2132 538.2288	C ₁₇ H ₂₄ O ₁₁ Na C ₂₆ H ₃₅ NO ₁₁ C ₂₆ H ₃₆ NO ₁₁	L-phenylalaninoscologanin
7	4.65	[M-H] ⁻	417.1186	C ₂₁ H ₂₁ O ₉	Liquiritin
8	4.77	[M-H] ⁻	549.1608	C ₂₆ H ₂₉ O ₁₃	4'-O-[β-D-apiofuranosyl (1 → 2)-β-glucopyranosyl]liquiritigenin
9	5.15	[M-H] ⁻	463.0877	C ₂₁ H ₁₉ O ₁₂	Quercetin-3-O-β-glucopyranoside
10	5.35	[M-H] ⁻ [M+H] ⁺	573.2084 575.2241	C ₂₈ H ₃₃ N ₂ O ₁₁ C ₂₈ H ₃₅ N ₂ O ₁₁	5-β-carboxystrictosidein
11	5.81	[M-H] ⁻	515.1190	C ₂₅ H ₂₃ O ₁₂	3,4-dicaffeoylquinic acid
12	6.02	[M-H] ⁻	515.1190	C ₂₅ H ₂₃ O ₁₂	3,5-dicaffeoylquinic acid
13	6.70	[M-H] ⁻ [M+Na] ⁺	757.2568 781.2531	C ₃₄ H ₄₅ O ₁₉ C ₃₄ H ₄₅ O ₁₉ Na	(<i>E</i>)-aldosecologanin
14	6.77	[M-H] ⁻	515.1190	C ₂₅ H ₂₃ O ₁₂	4,5-dicaffeoylquinic acid
15	7.26	[M-H] ⁻	783.2712	C ₃₆ H ₄₇ O ₁₉	Angoroside C
16	7.44	[M-H] ⁻	255.0657	C ₁₅ H ₁₁ O ₄	Liquiritigenin

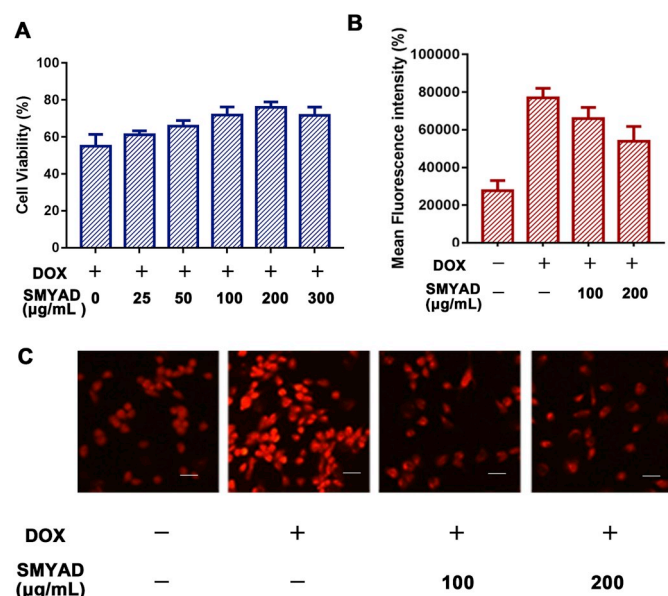


Fig. 6. Effect of SMYAD on DOX-induced oxidative damage in H9c2 cardiomyoblast cells. (A) The cyto-protection effect of SMYAD (25–300 μg/mL) was assessed in H9c2 cells and DOX (5 μM) treatment group. (B) Quantify the average fluorescence intensity of each group. Values (mean ± SD) are mean of three independent experiments. (C) Cardiomyocytes were treated with DOX in the presence or absence of 300 μg/mL SMYAD. Cardiomyocytes were identified by DCF-DA staining (red). Scale bars, 25 mm. Data are representative of three independent experiments (mean ± SD). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

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Appendix A. Supplementary data

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